SUMIT DUTTA

ACADEMIC PROFILE

Degree/Certificate	Institution	Percentage/CGPA	Year
IDD	Civil Engineering IIT (BHU), Varanasi	8.57	2027
CBSE (XII)	DPS, Ranchi	91.40	2022
CBSE (X)	DPS, Ranchi	94.00	2020

SKILLS

- **Languages:** C | C++ | Python | Javascript | HTML | CSS | SQL
- **AI/ML & Computer Vision:** Deep Learning | Computer Vision | Digital Image Processing | Image Segmentation | Pattern Recognition | PyTorch | OpenCV
- **Framework & Tools:** React.js, FastAPI, Docker, Git, MongoDB, MySQL, AWS

INTERNSHIP/TRAINING

Software Development Intern | Nation Innovation

May 2025 - July - 2025

- Contributed to the deployment of an **e-learning application** featuring **role-based dashboards**.
- Implemented **RESTful APIs** supporting full CRUD for users and courses, standardizing backend interactions.
- Developed a reusable **React** component library to standardize UI and accelerate feature delivery.
- Architected and optimized **MongoDB** schemas, improving query response time by **40%**.
- Implemented **Clerk** for secure, role-based authentication and route protection, enforcing access controls.
- **Exposure:** React, Node.js, Express, MongoDB, Clerk, RESTful APIs

PROJECTS

VitaminPScope – AI-Powered Digital Pathology Platform

- Built a viewer for **3+ whole-slide image formats** with interactive visualization using React and OpenSeadragon.
- Integrated **VitaminP**, a deep learning model trained on **14 datasets** across **34 cancer types**, for nuclei analysis.
- Enabled **interactive segmentation overlays** and **GeoJSON annotation export** for pathology analysis.
- **Exposure:** Python, PyTorch, Whole-slide imaging, computer vision, deep learning, REST APIs, and Docker.

AI Resume Builder & Mock Interview Platform | RAG

- Developed a **MERN AI interview platform** with **4 Langraph AI agents** for interviews and resume analysis.
- Designed **5+ Dockerized** microservices with **Redis caching** and **Firebase Authentication**.
- Developed a **React dashboard** with AI feedback, resume scoring, and personalized roadmaps.
  Exposure : React.js, TypeScript, REST API, Node.js, LangChain, Redis, PostgreSQL, Docker, Git, AWS

Compactify

A C++ utility implementing Huffman coding for lossless compression and accurate file restoration

- Implemented **Huffman coding** for lossless compression, reducing average file sizes by **30-40%**.
- Built an end-to-end **compression/decompression pipeline**, ensuring **100%** accuracy in file recovery.
- **Tech Stack:** C++ (STL), Priority Queue, Hash Map, Bitwise Operations.

POSITION OF RESPONSIBILITY

Tech Head | SHILP'25 (Departmental Fest)

Jan 2025 - April 2025

- Developed and deployed a **full-stack website** for SHILP 2024-25, including **registration** and **payment portal**.
- Built **scalable frontend/backend system** for **scheduling, authentication** and **secure transactions**.
- **Led the tech team** to integrate platforms with fest operations, ensuring **seamless execution** across 20+ events.

HONOURS AND ACHIEVEMENTS

- **Specialist** at **Codeforces** | Max Rating: **1470** | Profile Link
- **Knight** at **Leetcode** | Max Rating: **2025** | Profile Link
- **National Semi-finalist** at **Flipkart GRiD 7.0 Hackathon** out of over **1.6 Lakhs participants**.
- **Global Rank 856** in **Codeforces Round 856 (Div 2)** | **Global Rank 1191** in **Codeforces Round 1025 (Div 2)**.
- **Global Rank 533** in **Biweekly Contest 186** | **Global Rank 672** in **Biweekly Contest 184**.

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